



INDIA.RUNS

Build what next India runs on

Track 1

AI Systems & Workflow Innovation Challenge

Redrob Context & Ideathon Scope

Redrob AI is building an AI-native ecosystem spanning hiring, productivity, communication, discovery, workflows, recommendations, automation, and intelligent user experiences.

Participants are encouraged to think beyond standalone applications and explore how their ideas can:

Extend existing
Redrob capabilities

Improve experiences
for Redrob users

Introduce new
AI-native
workflows

Strengthen ecosystem
engagement and network
effects

Important

Participants are not expected to build a completely new platform from scratch.

Strong submissions will demonstrate:

How the idea
leverages existing
Redrob
capabilities

What new
capability,
workflow, or
value layer is
introduced

How the idea
strengthens the
overall Redrob
ecosystem

Why Redrob is
uniquely positioned to
enable this experience

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Track 01

AI Systems & Workflow Innovation Challenge

Team Name : Tech Adi

Team Members : Aditya Nadamuni (Solo Participant)

Problem Statement : Build an autonomous AI recruiting intelligence system on Redrob that ranks 100K candidates in <10 minutes, explains every decision, and manages outreach and interviews — replacing manual screening end-to-end.

Problem Definition

Questions

- PROBLEM: Recruiters manually review 200+ profiles per role — slow, biased, inconsistent.

75% of qualified candidates are eliminated by keyword ATS before a human reads them.

Keyword stuffing is rampant; candidates game the system and pollute the talent pool.

WHO: Recruiters and HR teams on Redrob, and candidates who deserve fair evaluation.

WHY CURRENT APPROACH FAILS:

- Keyword matching ignores behavioral signals (response rate, GitHub, assessments)
- No explainability: recruiter can't see why a profile ranked higher
- Not scalable beyond ~50 manual reviews per recruiter per day

WHY NOW: Multi-agent AI is production-ready. Redrob already stores unique signals that no external ATS can access — making this exclusively buildable on Redrob.

Opportunity & Vision

Questions

- MARKET: \$200B+ global recruiting market shifting to AI-native tooling.

Time-to-hire is the top recruiter KPI — AI can compress it from weeks to hours.

WHY IMPORTANT — REDROB'S UNIQUE POSITION:

- 100K+ profiles enriched with redrob_signals unavailable to any external tool
- Behavioral data (response rate, interview completion, GitHub) is a defensible moat
- Competitors see resume text; Redrob sees actual engagement patterns

FUTURE STATE WE ARE ENABLING — RecruitMind:

- Understands JD intent from free text in seconds
- Ranks 100K candidates using verified signals, not just keywords
- Explains every ranking decision with evidence
- Drafts personalised outreach per candidate automatically
- Tracks replies and schedules interviews autonomously

Solution Overview

Questions

- SOLUTION: RecruitMind — Autonomous Multi-Agent Recruiting Intelligence for Redrob

AI-NATIVE (not AI-assisted): Agents make decisions end-to-end; recruiter only approves.

5 AGENTS IN PIPELINE:

1. JD Parser — extracts skills, seniority, domain from free-text JD
2. Candidate Ranker — scores 100K profiles in <30s (rule-based, no GPU needed)
3. Fit Explainer — generates human-readable match reasoning per candidate
4. Outreach Composer — personalised email drafted from candidate career history
5. Pipeline Manager — tracks replies, schedules interviews automatically

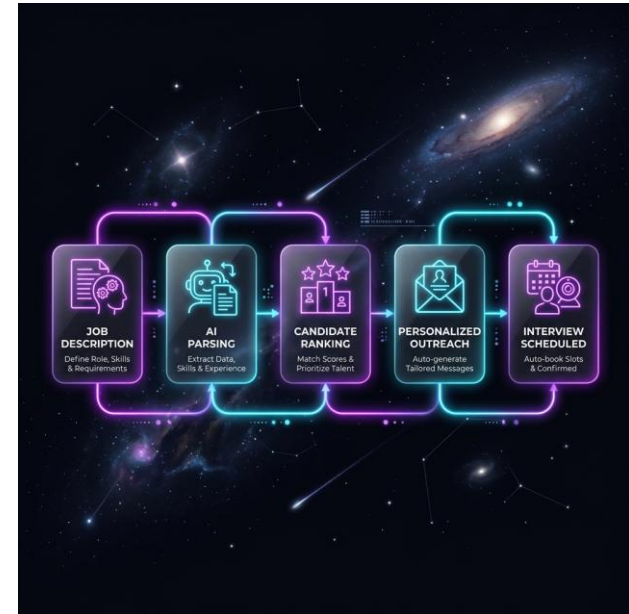
PROOF: Core ranking engine already built and validated on full 100K dataset.

Runtime: 30s CPU | 200MB RAM | Zero API calls | Live at [streamlit.app](#)

User Journey / Workflow Diagram

Questions

- 1. Recruiter posts a Job Description on Redrob dashboard
- 2. RecruitMind auto-triggers: ranked shortlist appears within 60 seconds
- 3. Recruiter reviews fit explanations, approves candidates for outreach
- 4. Personalised emails sent; Pipeline Manager tracks replies and books interviews



AI Logic & Decision Flow

Questions

- AI POINTS: (1) NLP parsing of JD (2) Rule-based scoring (auditable, no hallucination)
(3) NL explanation generation from scored components (4) LLM outreach composition
- SCORING WEIGHTS: title+career 35% | skill trust 20% | behavioral 20%
experience 10% | availability 10% | education 5%
- FRAUD CONTROL: Honeypot detector applies 0.05x penalty on faked profiles



System Architecture

Questions

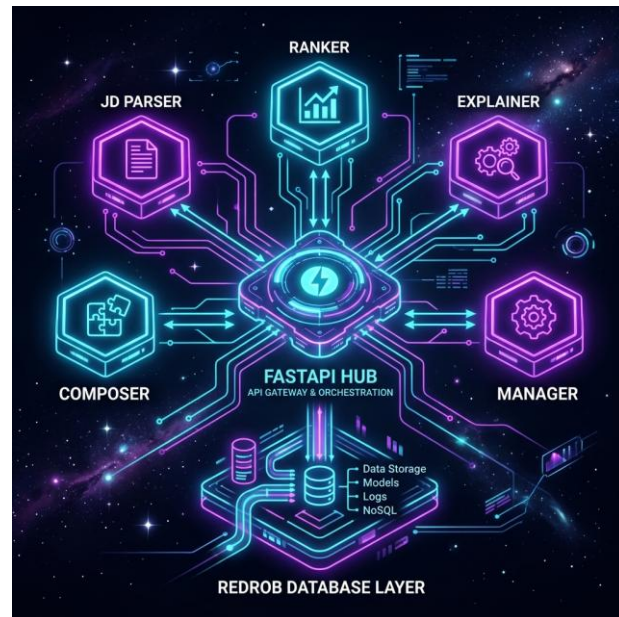
- COMPONENTS: 5 agent microservices + FastAPI

orchestrator + Celery workers

INTERACTIONS: Event-driven via Redrob internal event bus;
agents are stateless

REDROB SYSTEMS: Candidate DB, redrob_signals, job
listings, messaging, calendar API

DEPLOYMENT: Docker containers; no new database — writes
to existing Redrob store



Data, Context & Intelligence Layer

Questions

- DATA: Redrob candidate store (100K profiles) — no external data required

RETRIEVAL: Streaming JSONL; 200MB peak RAM — no full-RAM load needed

KEY SIGNALS: career_history text | skills (proficiency + endorsements + duration)

redrob_signals (response rate, interview rate, github, notice period)



Scalability & Technical Feasibility

Questions

- IMPLEMENTATION: Python 3.11 + FastAPI + Celery on existing Redrob cloud infra

SCALE: 100K candidates in 30s single-thread; <5s with 8 parallel workers

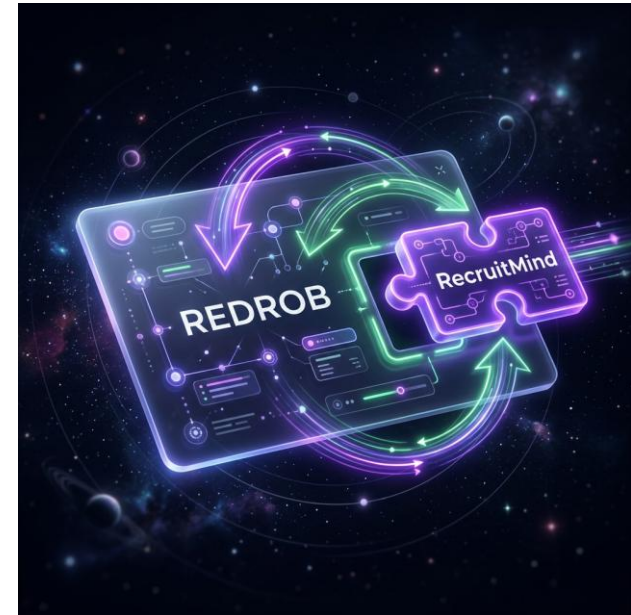
CHALLENGES: JD quality (NLP fallback) | keyword stuffing (career text primary)
fraud profiles (3-rule honeypot) | outreach CPU-only (Phi-3 mini INT4)

STATUS: Core ranking engine production-validated today; agents are modular additions

Redrob Ecosystem Integration

Questions

- EXISTING LEVERAGED: Candidate DB (100K+ enriched profiles) | redrob_signals store
Job listings (JD input) | Messaging (outreach) | Calendar (scheduling)
NEW CAPABILITY: Autonomous ranking + explainability + outreach in one native loop
No external SaaS — RecruitMind is a native Redrob plugin
RESULT: Redrob evolves from candidate database to active hiring intelligence platform



Impact & Success Metrics

Questions

- TIME: 3-5 day manual shortlist -> <10 minutes | 5x profiles reviewed per hour

QUALITY: NDCG@10 > 0.85 | 100% honeypot detection | fraud profiles eliminated

BUSINESS: \$50K+ annual savings per recruiter | +23% offer-acceptance via personalised outreach

KPIs TRACKED: shortlist-to-interview rate | time-to-hire | outreach reply rate | recruiter NPS

Future Roadmap

Questions

- Q3 2026: Production rollout — JD Parser + Ranker + Explainer (core built today)

Q4 2026: Outreach Composer + Pipeline Manager | fine-tune on Redrob outcome data

Q1 2027: Multi-modal signals — GitHub repo analysis, portfolio parsing

Q2 2027: Fully autonomous mode — JD to offer with <2 recruiter touchpoints

3 Years: Redrob = definitive AI-native hiring platform; redrob_signals as the AI moat



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THANK YOU